

TID Marital Status Cheat

1	Single	No
2	Single	No
3	Single	Yes
4	Married	No
5	Married	No
6	Married	No
7	Married	Yes
8	Divorced	No
9	Divorced	Yes
10	Divorced	Yes

Step 1: Initialize n_i for each class

Line 1–2:

- **Classes:** $c_1 = \text{No}$, $c_2 = \text{Yes}$
- **Initially:** $n_{\text{No}} = 0$, $n_{\text{Yes}} = 0$

Step 2: Initialize $n_{v,i}$ for each category–class combination

Line 3:

- **Domain(X)** = {Single, Married, Divorced}
- Start all counts at zero:

Step-3: Loop Lines 4-5

Marital Status n_v , No n_v , Yes Total

Single	2	1	3
Married	3	1	4
Divorced	1	2	3
Total	6	4	10

Step 4: Initialize best split tracking

Line 6:

$V^* = \emptyset$, $\text{score}^* = 0$

Step 5: Generate candidate subsets V

Line 7: For $l = 1$, candidates are:

1. {Single}
2. {Married}
3. {Divorced}

(If $l > 1$, we'd also check combinations like {Single, Married}, {Single, Divorced}, {Married, Divorced}.)

Step 6: For each candidate V, compute probabilities

Example: $V = \{\text{Divorced}\}$

- $DY =$ points where Marital Status $\in \{\text{Divorced}\}$
 - Yes count = 2
 - No count = 1
 - $\hat{P}(\text{Yes} \mid DY) = 2/3, \quad \hat{P}(\text{No} \mid DY) = 1/3$
- $DN =$ points where Marital Status $\notin \{\text{Divorced}\}$
 - Yes count = $4 - 2 = 2$
 - No count = $6 - 1 = 5$
 - $\hat{P}(\text{Yes} \mid DN) = 2/7, \quad \hat{P}(\text{No} \mid DN) = 5/7$

Step 7: Calculate Gain

Entropy of full dataset:

$$H(D) = -\frac{6}{10} \log_2 \frac{6}{10} - \frac{4}{10} \log_2 \frac{4}{10} \approx 0.971$$

Entropy of DY:

$$H(DY) \approx 0.918$$

Entropy of DN:

$$H(DN) \approx 0.863$$

Gain:

$$\text{Gain} \approx 0.971 - \frac{3}{10}(0.918) - \frac{7}{10}(0.863) \approx 0.091$$

Step 8: Update best split if needed

If $\text{score}(V) > \text{score}^*$ then set $V^* = V, \text{score}^* = \text{score}(V)$.

For $V = \{\text{Divorced}\}$, $\text{gain} \approx 0.091 \rightarrow$ currently best.

Step 9: Repeat for other candidates

- $V = \{\text{Married}\} \rightarrow \text{gain} \approx 0.046$
- $V = \{\text{Single}\} \rightarrow \text{gain} \approx 0.006$

Step 10: Return best split

Line 14: